

Сетевые технологии

Лабораторная работа №8

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Цель данной работы — изучение принципов маршрутизации в IPv4- и IPv6-сетях и принципов настройки сетевого оборудования.

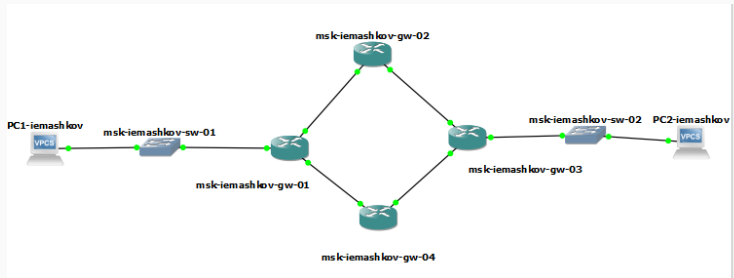


Рис. 1: Топология

```
frr# configure terminal
frr(config)# hostname msk-iemashkov-gw-01
msk-iemashkov-gw-01(config)# exit
msk-iemashkov-gw-01# configure terminal
msk-iemashkov-gw-01(config)# router rip
msk-iemashkov-gw-01(config-router)# version 2
msk-iemashkov-gw-01(config-router)# network eth0
msk-iemashkov-gw-01(config-router)# network eth1
msk-iemashkov-gw-01(config-router)# network eth2
msk-iemashkov-gw-01(config-router)# exit
msk-iemashkov-gw-01(config)# exit
msk-iemashkov-gw-01# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-iemashkov-gw-01#
```

Рис. 2: Настройка rip

```
msk-iemashkov-gw-02# configure terminal
msk-iemashkov-gw-02(config)# router rip
msk-iemashkov-gw-02(config-router)# version 2
msk-iemashkov-gw-02(config-router)# network eth0
msk-iemashkov-gw-02(config-router)# network eth1
msk-iemashkov-gw-02(config-router)# exit
msk-iemashkov-gw-02(config)# exit
msk-iemashkov-gw-02# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-iemashkov-gw-02#
```

Рис. 3: Настройка rip

```
frr# configure terminal
frr(config)# hostname msk-iemashkov-gw-03
msk-iemashkov-gw-03(config)# exit
msk-iemashkov-gw-03# configure terminal
msk-iemashkov-gw-03(config)# router rip
msk-iemashkov-gw-03(config-router)# version 2
msk-iemashkov-gw-03(config-router)# network eth0
msk-iemashkov-gw-03(config-router)# network eth1
msk-iemashkov-gw-03(config-router)# network eth2
msk-iemashkov-gw-03(config-router)# exit
msk-iemashkov-gw-03(config)# exit
msk-iemashkov-gw-03# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-iemashkov-gw-03#
```

Рис. 4: Настройка rip

```
msk-iemashkov-gw-04# configure terminal
msk-iemashkov-gw-04(config)# router rip
msk-iemashkov-gw-04(config-router)# version 2
msk-iemashkov-gw-04(config-router)# network eth0
msk-iemashkov-gw-04(config-router)# network eth1
msk-iemashkov-gw-04(config-router)# exit
msk-iemashkov-gw-04(config)# exit
msk-iemashkov-gw-04# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-iemashkov-gw-04#
```

Рис. 5: Настройка rip

Выполнение лабораторной работы

[illegible]

Рис. 6: show ip route

Выполнение лабораторной работы

[illegible]

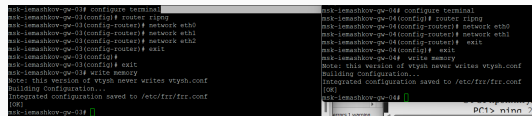
Рис. 7: Взаимный пинг



```
nsk-lemashkov-gw-01# configure terminal
nsk-lemashkov-gw-01(config)# router ipng
nsk-lemashkov-gw-01(config-router)# network eth0
nsk-lemashkov-gw-01(config-router)# network eth1
nsk-lemashkov-gw-01(config-router)# network eth2
nsk-lemashkov-gw-01(config-router)# exit
nsk-lemashkov-gw-01(config)# exit
nsk-lemashkov-gw-01# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated Configuration saved to /etc/frr/frr.conf
[OK]
nsk-lemashkov-gw-01#

nsk-lemashkov-gw-02(config-if)# exit
nsk-lemashkov-gw-02(config)# router ipng
nsk-lemashkov-gw-02(config-router)#
nsk-lemashkov-gw-02(config-router)# network eth0
nsk-lemashkov-gw-02(config-router)# network eth1
nsk-lemashkov-gw-02(config-router)# exit
nsk-lemashkov-gw-02(config)# exit
nsk-lemashkov-gw-02# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated Configuration saved to /etc/frr/frr.conf
[OK]
nsk-lemashkov-gw-02#
```

Рис. 8: Настройка ipng



```
msk-lemashkov-gw-03# configure terminal
msk-lemashkov-gw-03(config)# router ripng
msk-lemashkov-gw-03(config-router)# network eth0
msk-lemashkov-gw-03(config-router)# network eth1
msk-lemashkov-gw-03(config-router)# network eth2
msk-lemashkov-gw-03(config-router)# exit
msk-lemashkov-gw-03(config)#
msk-lemashkov-gw-03(config)# exit
msk-lemashkov-gw-03# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/irrr/irrr.conf
[OK]
msk-lemashkov-gw-03#

msk-lemashkov-gw-04# configure terminal
msk-lemashkov-gw-04(config)# router ripng
msk-lemashkov-gw-04(config-router)# network eth0
msk-lemashkov-gw-04(config-router)# network eth1
msk-lemashkov-gw-04(config-router)# exit
msk-lemashkov-gw-04(config)#
msk-lemashkov-gw-04(config)# exit
msk-lemashkov-gw-04# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/irrr/irrr.conf
[OK]
msk-lemashkov-gw-04#
```

Рис. 9: Настройка ripng

```
VPCS> ping 2001:11::a

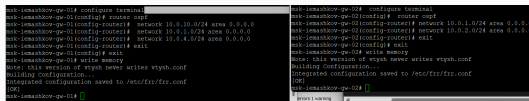
2001:11::a icmp6_seq=1 ttl=58 time=6.256 ms
2001:11::a icmp6_seq=2 ttl=58 time=4.867 ms
2001:11::a icmp6_seq=3 ttl=58 time=4.583 ms
2001:11::a icmp6_seq=4 ttl=58 time=4.787 ms
2001:11::a icmp6_seq=5 ttl=58 time=7.093 ms

VPCS> trace 2001:11::a

trace to 2001:11::a, 64 hops max
 1 2001:10::1   1.541 ms   1.010 ms   1.194 ms
 2 2001:1::2    2.936 ms   2.341 ms   2.109 ms
 3 2001:2::2    3.891 ms   4.286 ms   2.977 ms
 4 2001:11::a   3.306 ms   4.030 ms   4.863 ms

VPCS> 
```

Рис. 10: Пинг



The image shows two terminal windows side-by-side, both running Mikrotik WinBox. The left window is for router 'mik-iamashkov-qw-01' and the right window is for 'mik-iamashkov-qw-02'. Both show the configuration of OSPF (OSPFv2) on their respective interfaces. The configuration includes setting the OSPF process name to 'ospf', enabling it on the interface, and defining the network and area. The left router has interfaces 10.0.10.0/24 and 10.0.1.0/24 in area 0.0.0.0. The right router has interfaces 10.0.1.0/24 and 10.0.2.0/24 in area 0.0.0.0. Both configurations are saved to the file '/etc/frr/frr.conf'.

```
mik-iamashkov-qw-01# configure terminal
mik-iamashkov-qw-01(config)# router ospf
mik-iamashkov-qw-01(config-router)# network 10.0.10.0/24 area 0.0.0.0
mik-iamashkov-qw-01(config-router)# network 10.0.1.0/24 area 0.0.0.0
mik-iamashkov-qw-01(config-router)# exit
mik-iamashkov-qw-01(config)# exit
mik-iamashkov-qw-01# write memory
Note: this version of vysh now writes vysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
mik-iamashkov-qw-01#
```

```
mik-iamashkov-qw-02# configure terminal
mik-iamashkov-qw-02(config)# router ospf
mik-iamashkov-qw-02(config-router)# network 10.0.1.0/24 area 0.0.0.0
mik-iamashkov-qw-02(config-router)# network 10.0.2.0/24 area 0.0.0.0
mik-iamashkov-qw-02(config-router)# exit
mik-iamashkov-qw-02(config)# exit
mik-iamashkov-qw-02# write memory
Note: this version of vysh now writes vysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
mik-iamashkov-qw-02#
```

Рис. 11: Настройка ospf

```
mk-lamashkov-gw-03# configure terminal
mk-lamashkov-gw-03(config)# router ospf
mk-lamashkov-gw-03(config-router)# network 10.0.11.0/24 area 0.0.0.0
mk-lamashkov-gw-03(config-router)# network 10.0.2.0/24 area 0.0.0.0
mk-lamashkov-gw-03(config-router)# network 10.0.3.0/24 area 0.0.0.0
mk-lamashkov-gw-03(config-router)# exit
mk-lamashkov-gw-03(config)# exit
mk-lamashkov-gw-03# write memory
Note: this version of viosdpx never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/tr/tr.conf
[OK]
mk-lamashkov-gw-03#

mk-lamashkov-gw-04# configure terminal
mk-lamashkov-gw-04(config)# router ospf
mk-lamashkov-gw-04(config-router)# network 10.0.3.0/24 area 0.0.0.0
mk-lamashkov-gw-04(config-router)# network 10.0.4.0/24 area 0.0.0.0
mk-lamashkov-gw-04(config-router)# exit
mk-lamashkov-gw-04(config)# exit
mk-lamashkov-gw-04# write memory
Note: this version of viosdpx never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/tr/tr.conf
[OK]
mk-lamashkov-gw-04#
```

Рис. 12: Настройка ospf

В ходе данной лабораторной работы мной были изучены принципы маршрутизации в IPv4- и IPv6-сетях и принципы настройки сетевого оборудования.